Upland Ecology

<u>Field Report</u>

Castle Head, FSC

A broad assessment of Upland Ecology in Cumbria

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Abstract

This report presents a broad assessment of upland ecosystems in Cumbria. We evaluated plant communities in raised bogs, calcareous and acidic grasslands, and limestone pavements, along with freshwater and terrestrial invertebrate assemblages, using field surveys and statistical analyses. Vegetation was sampled via quadrats to calculate diversity indices and community similarity (e.g. Shannon and Sørensen). Key findings include markedly higher plant diversity and evenness in the calcareous grassland at Scout Scar (H'=3.03, E=0.93) compared to the acidic grassland at Gummer's How (H'=2.12, E=0.80). Insects were surveyed using pitfall traps, revealing significantly larger ground beetles in woodland than grassland ($p < 0.001$) and greater beetle species richness in grassland (16 vs. 8 species). Freshwater sampling showed distinct invertebrate communities between lake (lentic) and stream (lotic) sites. Overall, these results highlight that substrate chemistry, moisture regimes, and habitat structure are primary determinants of species composition and biodiversity in these upland landscapes.

Section 1: Raised Bog plant communities in Meathop Moss

A. Distribution of plant species found at Meathop Moss

Raised bogs develop through the gradual accumulation of peat, which results from the limited decomposition of sphagnum mosses and other vegetation under waterlogged, anaerobic conditions. Variations in plant growth and decomposition contribute to uneven surface topography, which is further shaped by patterns of plant distribution and weathering processes. (Van de Molen, 1994) This leads to the formation of elevated structures known as hummocks and lower, depressed areas referred to as hollows. Hummocks are situated above the water-table and are subject to fluctuating moisture levels, nutrient leaching, and occasional periods of desiccation. In contrast, hollows remain persistently waterlogged, characterised by low oxygen availability and elevated concentrations of dissolved nutrients. Observations from Meathop Moss demonstrate the distribution of plant species across these distinct microclimates.

Microclimate Type:	*Hollow*		*Hummock*	
Scientific name	**Frequency by scientific name**	**Median Domin Score**	**Frequency by scientific name**	**Median Domin Score**
Andromeda polifolia	13	1	15	2
Betula pendula	N/A	N/A	1	1
Betula pubescens	N/A	N/A	6	1
Calluna vulgaris	23	4	24	6.5
Cladonia arbuscula	4	6	1	2
Cladonia portentosa	6	2.5	8	5.5
Drosera rotundifolia	6	2	N/A	N/A
Empetrum nigrum	N/A	N/A	1	6
Erica tetralix	11	5	19	4
Eriophorum angustifolium	10	2.5	10	2.5
Eriophorum vaginatum	27	6	22	7.5
Hypnum jutlandicum	1	2	N/A	N/A
Molinia caerulea	6	3.5	2	2
Narthecium ossifragum	13	4	7	5
Pinus sylvestris	N/A	N/A	1	1
Polytrichum commune	3	4	7	7
Sphagnum capillifolium	24	8.5	15	7
Sphagnum papillosum	14	4	7	4
Sphagnum subnitens	5	9	5	8
Trichophorum germanicum	3	7	4	7
Vaccinium myrtillus	N/A	N/A	4	2.5
Vaccinium oxycoccos	25	3	26	4

Table 1: Frequency and median Domin value of plant species observed at Meathop Moss through random sampling using 50x50cm quadrats, displaying differences in plant community composition in hollow and hummock microclimates.

B. Comparison of hummock and hollow plant communities

Hummocks

Scientific Name	Frequency
Vaccinium oxycoccos	26
Calluna vulgaris	24
Eriophorum vaginatum	22
Erica tetralix	19
Andromeda polifolia	15
Sphagnum capillifolium	15
Eriophorum angustifolium	10
Cladonia portentosa	8
Narthecium ossifragum	7
Sphagnum papillosum	7
Polytrichum commune	7
Betula pubescens	6
Sphagnum subnitens	5
Trichophorum germanicum	4
Vaccinium myrtillus	4
Molinia caerulea	2
Cladonia arbuscula	1
Empetrum nigrum	1
Pinus sylvestris	1
Betula pendula	1

Hollows

Scientific Name	Frequency
Eriophorum vaginatum	27
Vaccinium oxycoccos	25
Sphagnum capillifolium	24
Calluna vulgaris	23
Sphagnum papillosum	14
Narthecium ossifragum	13
Andromeda polifolia	13
Erica tetralix	11
Eriophorum angustifolium	10
Cladonia portentosa	6
Drosera rotundifolia	6
Molinia caerulea	6
Sphagnum subnitens	5
Cladonia arbuscula	4
Trichophorum germanicum	3
Polytrichum commune	3
Hypnum jutlandicum	1

Table 2: Species listed by abundance in hummocks and hollows at Meathop Moss

In hollows, the most frequent species were *Eriophorum vaginatum* (27 occurrences), *Sphagnum capillifolium* (24), *Vaccinium oxycoccos* (25), *Calluna vulgaris* (23), and *Sphagnum papillosum* (14). Of these, *E. vaginatum* and *S. capillifolium* had particularly high median Domin scores (6 and 8.5 respectively), indicating not only frequent presence but also local dominance.

In hummocks, the most frequent species were *Vaccinium oxycoccos* (26 occurrences), *Calluna vulgaris* (24), *Eriophorum vaginatum* (22), *Erica tetralix* (19), and *Andromeda polifolia* (15, equal with *S. capillifolium*). *Calluna* and *Eriophorum* both had relatively high cover (median Domin 6.5 and 7.5 respectively), while *Erica tetralix* was present at moderate cover (4).

The hollow community was characterised by species adapted to persistently wet, waterlogged microhabitats. *Eriophorum vaginatum* forms tussocks but can also colonise hollows, tolerating high water tables and anaerobic soils. *Sphagnum capillifolium* and *S. papillosum* are classic peat-forming mosses whose morphology enables water retention, allowing them to dominate low-lying saturated areas.

In contrast, hummocks supported more ericaceous shrubs alongside *Eriophorum*. *Calluna vulgaris* and *Erica tetralix* are better suited to drier, aerated conditions, their lignified leaves reducing desiccation stress. *Andromeda polifolia* is also typical of slightly raised, acidic peat, and its frequency reflects its affinity for these microhabitats. Interestingly, *Vaccinium oxycoccos* was common in both hollows and hummocks, suggesting ecological flexibility, though its trailing growth habit may allow it to exploit niches across the bog surface.

Together, these patterns support the classic microtopographic model of bog vegetation: hollows dominated by graminoids and *Sphagnum* mosses tolerant of waterlogging, and hummocks dominated by dwarf shrubs that avoid prolonged inundation. (Edgell, 1969)

Section Two: Grasslands; Calcareous and Acidic

A. Analysis of communities using the Shannon Diversity & Pielou Evenness Indices

The plant communities at Scout Scar and Gummer's How are associated with differing geologies, with limestone producing calcareous soils at Scout Scar and siltstone forming acidic soils at Gummer's How. To quantify compositional differences between these sites, statistical indices were employed.

The Shannon Diversity Index (H') estimates species diversity within a community by combining species richness (S) and the relative abundances of each species. The Pielou Evenness Index (E) expresses how evenly species cover is distributed, producing a value between 0 (no evenness) and 1 (maximum evenness).

For the calcareous grassland at Scout Scar, the calculated indices were H' = **3.03**, S = **26**, and E = **0.93**. For the acidic grassland at Gummer's How, the values were H' = **2.12**, S = **14**, and E = **0.80**. These statistics quantitatively describe the observed community structure; the higher values of H', S, and E, at Scout Scar indicate greater species diversity and a more even distribution of cover among species compared to Gummer's How.

	scientific name	**Percent Cover**	**pi**	**ln(pi)**	**pi × ln(pi)**
1	Agrostis capillaris	184	0.1925	-1.6478	-0.3172
2	Calluna vulgaris	4	0.0042	-5.4765	-0.0229
3	Deschampsia flexuosa	24	0.0251	-3.6847	-0.0925
4	Dicranum scoparium	10	0.0105	-4.5602	-0.0477
5	Festuca ovina	264	0.2762	-1.2868	-0.3554
6	Galium saxatile	118	0.1234	-2.0921	-0.2582
7	Hypnum jutlandicum	8	0.0084	-4.7833	-0.0400
8	Luzula multiflora	34	0.0356	-3.3364	-0.1187
9	Nardus stricta	66	0.0690	-2.6731	-0.1845
10	Polytrichastrum commune	12	0.0126	-4.3779	-0.0550
11	Potentilla erecta	32	0.0335	-3.3970	-0.1137
12	Rhytidiadelphus squarrosus	50	0.0523	-2.9507	-0.1543

13	Rumex acetosella	24	0.0251	-3.6847	-0.0925
14	Vaccinium myrtillus	124	0.1297	-2.0425	-0.2649
15	bare ground	2	0.0021	-6.1696	-0.0129
	H' (- Σ pi*lm(pi)) = 2.118				

Table 3: Shows data and equation for the Shannon Diversity Index (H'), for point quadrat data collected from Gummer's How. "Bare ground" was excluded from calculations

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Pielou Evenness Index (E), Gummer's How

(E = H'/ lnS)

H' = 2.118

S = 14

lnS = 2.639

E = 2.118 / 2.639

E = 0.803

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	scientific name	Percent Cover	pi	ln_pi	pi_ln_pi
1	Achillea millefolium	25	0.023474178	-3.751854	-0.08807170
2	Briza media	14	0.013145540	-4.331673	-0.05694218
3	Calluna vulgaris	14	0.013145540	-4.331673	-0.05694218
4	Carex caryophyllea	34	0.031924883	-3.444370	-0.10996109
5	Carex flacca	119	0.111737089	-2.191607	-0.24488374
6	Carex pulicaris	2	0.001877934	-6.277583	-0.01178889
7	Ctenidium molluscum	18	0.016901408	-4.080358	-0.06896380
8	Dicranum scoparium	12	0.011267606	-4.485823	-0.05054449
9	Ditrichum gracile	2	0.001877934	-6.277583	-0.01178889
10	Festuca ovina	152	0.142723005	-1.946850	-0.27786022
11	Galium sterneri	16	0.015023474	-4.198141	-0.06307067
12	Galium verum	14	0.013145540	-4.331673	-0.05694218
13	Helianthemum nummularium	34	0.031924883	-3.444370	-0.10996109
14	Helictotrichon pratense	9	0.008450704	-4.773506	-0.04033948
15	Leontodon hispidus	4	0.003755869	-5.584436	-0.02097441
16	Linum catharticum	20	0.018779343	-3.974998	-0.07464785
17	Lotus corniculatus	107	0.100469484	-2.297901	-0.23086895
18	Pilosella officinarum	6	0.005633803	-5.178971	-0.02917730
19	Plantago lanceolata	72	0.067605634	-2.694064	-0.18213390
20	Potentilla erecta	60	0.056338028	-2.876386	-0.16204989
21	Poterium sanguisorba	35	0.032863850	-3.415382	-0.11224260
22	Pseudoscleropodium purum	22	0.020657277	-3.879688	-0.08014378

23	Rhytidiadelphus squarrosus	14	0.013145540	-4.331673	-0.05694218
24	Sesleria caerulea	151	0.141784038	-1.953450	-0.27696806
25	Thymus polytrichus	105	0.098591549	-2.316770	-0.22841392
26	Viola lutea	2	0.001877934	-6.277583	-0.01178889
27	bare ground	2	0.001877934	-6.277583	-0.01178889
	H' (- Σ pi*lm(pi)) = 3.031				

Table 4: Shows data and equation for the Shannon Diversity Index (H'), for point quadrat data collected from Scout's Scar. "Bare ground" was excluded from calculations

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Pielou Evenness Index (E), Scout's Scar

(E = H'/ lnS)

H' = 3.031

S = 26

lnS = 3.258

E = 3.031 / 3.258

E = 0.931

B. Comparison of Scout's Scar & Gummer's How using the Sørensen Index

The Sørensen Similarity Index compares the communities of two sites and returns a value between 0 (no species the same) and 1 (all species the same). The result of 0.33 indicates low similarity between the sites, with approximately one third of species shared.

		Gummer's How	Scout's Scar	Combined
1	No. Species	16	32	40
2	No. Unique species	8	24	32
3	No. Shared species			8

{Sørensen Index = 2 × shared species / (Gummer's How species + Scout's Scar species)

= 2 × 8 / (16 + 32)

= 16 / 48

Sørensen Index = 0.33}

Table 5: Calculation of Sorensen similarity index using 50x50cm quadrat data from Scout's Scar and Gummer's How

C. Rank abundance plots for Gummer's How & Scout's Scar

Rank abundance plots show the distribution of total cover among species in the acidic grassland at Gummer's How and calcareous grassland at Scout's Scar. The results reflect the differences in community structure observed in the Shannon Diversity and Pielou Evenness indices.

In Gummer's How, the community was dominated by a few species: the top three, ***Festuca ovina*** (264), ***Agrostis capillaris*** (184), and ***Vaccinium myrtillus*** (124), accounted for 68.6% of the total vegetation cover (572 out of 834 total cover units). In contrast, the top three species at Scout Scar—***Festuca ovina*** (152), ***Sesleria caerulea*** (151), and ***Carex flacca*** (119)—represented 44.7% of the total cover (422 out of 944 total cover units). This indicates a more even spread of abundance in the calcareous grassland compared to the acidic grassland.

Scout Scar also showed a slower tapering of abundance among less common species, suggesting a more balanced community structure. Both sites shared ***Festuca ovina*** as the most dominant species, but only ***Potentilla erecta*** appeared in the top seven ranks for both communities.

These rank abundance results support the interpretation that Gummer's How was dominated by a small number of species, while Scout Scar displayed greater species richness and evenness in cover distribution.

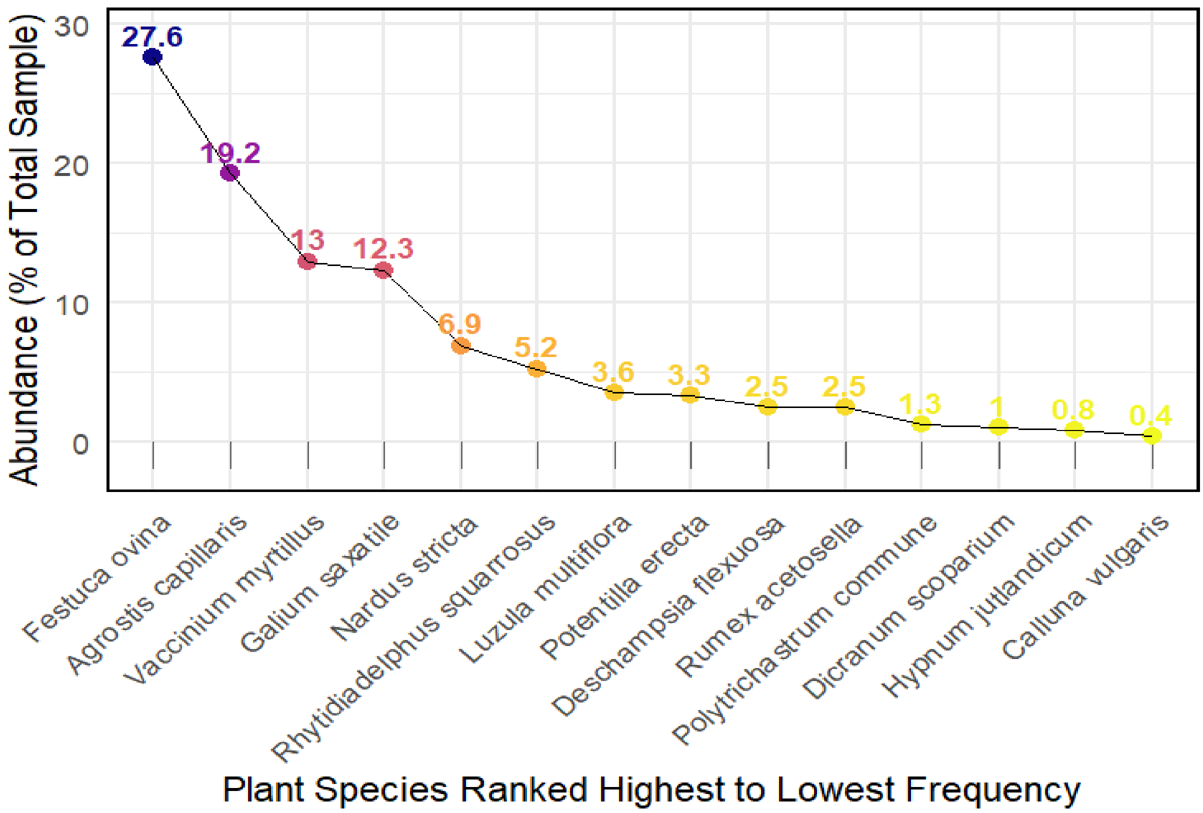

Figure 1: Abundance Rank Plot for (acidic) grassland community at Gummer's How from data sampled using 10-pint quadrats,

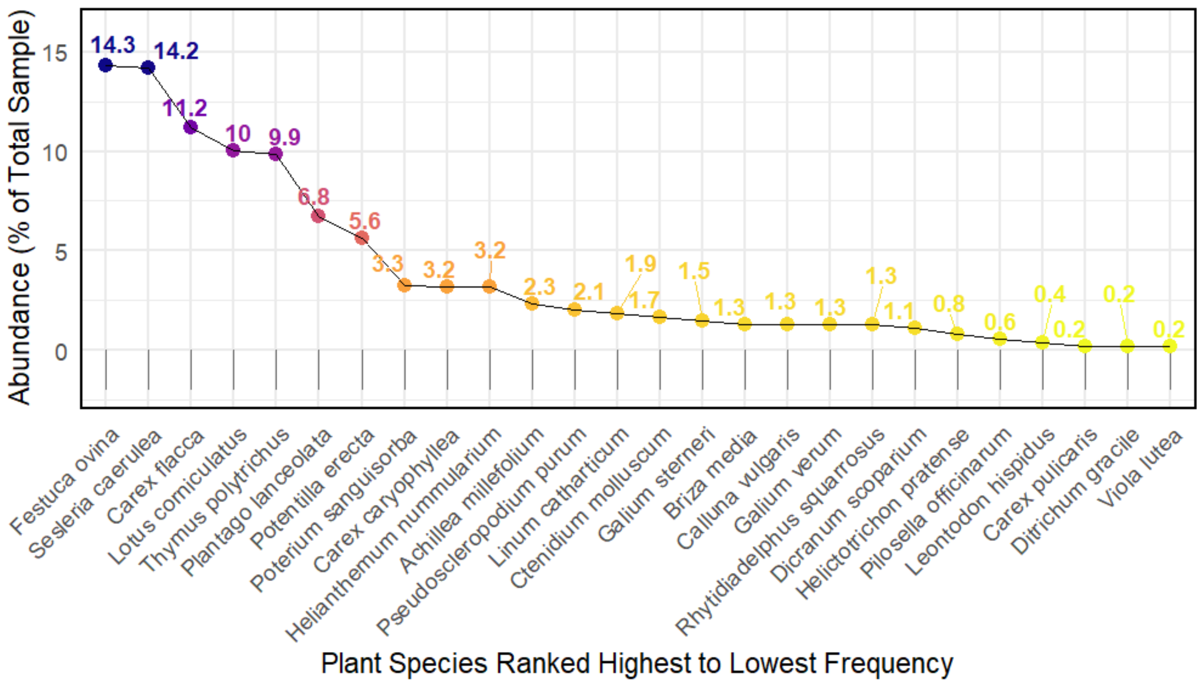

Figure 2: Abundance Rank Plot for the (calcareous) grassland community at Scout's Scar from data sampled using 10-point quadrats

D. Comparison of Scout's Scar & Castle Hill using the Sørensen Index

Castle Hill NNR, located on Cretaceous chalk in the South Downs National Park (Natural England, 2014), shares calcareous soils with Scout's Scar in Cumbria, which lies on Carboniferous limestone. Despite both sites supporting calcareous grassland communities, the Sørensen Index of **0.43** indicates substantial differences in plant species composition. This similarity is notably higher than that observed between Scout's Scar and the acidic soils of Gummer's How (Sørensen Index = **0.33**), highlighting the importance of soil chemistry in shaping plant communities.

		Castle Hill	Scout's Scar	Combined
1	No. Species	28	32	47
2	No. Unique species	19	15	34
3	No. Shared species			13

{Sørensen Index = 2 × shared species / (Calcareous Grassland species + Castle Hill species)

= 2 × 13 / (32 + 28)

= 26 / 60

Sørensen Index = 0.43}

Table 6: Calculation of Sorensen similarity index for Scout's Scar and Castle Hill with columns for each site and total distinct species

While soil type is a major factor, the relatively low Sørensen Index suggests other environmental gradients play a significant role. Climatic and geographic data (weatherandclimate.com, 2025) reveal that Scout's Scar (Cumbria) is cooler, wetter, and significantly higher in elevation than Castle Hill (East Sussex):

		Cumbria	East Sussex
1	Latitude	54.6°N	50.9°N
2	Elevation	348.7m	59.7m
3	Mean Annual Temp	8.0°C	11.8°C
4	Annual Precipitation	943.2mm/year	721.2mm/year
5	Rainy Days	162	132
6	Average Humidity	90%	82%

Table 7: Climate and geographic data comparing Cumbria and East Sussex

Sources: Weather and Climate (n.d.a; n.d.b)

These differences are reflected in the species found uniquely at each site. For example, **Castle Hill** hosts several southern, warmth-loving chalk grassland specialists such as *Anthyllis vulneraria* and *Scabiosa columbaria*, which favour drier, warmer conditions and are rare or absent in northern upland grasslands. In contrast, **Scout's Scar** supports northern and montane species adapted to cooler, wetter, and higher-altitude environments, including *Viola lutea* and the characteristic grass *Sesleria caerulea*. The presence of mosses such as *Dicranum scoparium* and *Rhytidiadelphus squarrosus* at Scout's Scar further indicates adaptation to higher humidity and rainfall.

Thus, while both sites are calcareous grasslands, climatic and geographic differences—such as temperature, rainfall, latitude, and elevation—shape their plant communities as much as, or more than, soil chemistry alone. This highlights the complex interplay of environmental factors driving biodiversity patterns in British grasslands.

E. Field experiment proposal for *Galium saxatile* & *Galium sterneri*

Galium saxatile and *G. sterneri* are morphologically very similar, differing mainly in the orientation of trichomes along the leaf margins (Streeter, 2010). Ecologically, however, *G. saxatile* is a calcifuge typical of acidic grasslands, while *G. sterneri* is a calcicole confined to calcareous soils. Their contrasting distributions suggest that environmental filtering by soil chemistry restricts each species to its home habitat.

Hypothesis

H_0: Both species perform equally well in acidic and calcareous soils.

H_1: Each species shows higher performance in its native soil type and reduced survival, growth, and reproduction when transplanted to the foreign soil.

A reciprocal transplant experiment could be used to test this hypothesis (Agren, et al., 2012). Two sites would be selected: one with acidic soils and one with calcareous soils. At each site, 1 × 1 m plots would be established, half sown with the native species (control) and half with the foreign species (transplant). Plots would be subdivided into 10 × 10 cm cells, each sown with a single seed to standardise starting conditions. Existing individuals would be removed prior to sowing.

Plots would be monitored weekly through one growing season. Variables measured to determine outcome would include germination success, growth rate, survival, and reproductive output (flowering and fruiting). Stress symptoms such as chlorosis, stunted growth, or necrosis would also be recorded.

If adaptations are habitat-specific, transplanted individuals should show reduced performance compared to natives, reflecting maladaptation to soil chemistry. Conversely, similar performance across soils would support the null hypothesis.

F. Challenges concerning plant growth in non-suitable soil pH

Calcifuge species are adapted to acidic soils, where calcium availability is low, but micronutrients such as iron and manganese are relatively abundant. When growing in calcareous conditions, however, these plants experience physiological stress: high calcium levels disrupt cellular processes, while the reduced availability of iron and phosphorus leads to nutrient deficiencies. This imbalance often results in symptoms such as chlorosis, stunted growth, and poor reproductive output In contrast, calcicole species are adapted to calcareous soils, where abundant base cations support growth but soluble metals are scarce. When transplanted into acidic soils, they lack the root exudate and uptake mechanisms necessary to mitigate the higher solubility of iron and manganese. As a result, they suffer from metal toxicity and impaired nutrient assimilation, leading to poor performance and developmental abnormalities (George et al., 2012).

These ecological specialisations reflect broader principles of root system regulation. As Yu and Hochholdinger (2023) emphasise, root architecture and physiology continuously adjust to environmental conditions through hormonal and metabolic pathways. However, the contrasting evolutionary trajectories of calcicole and calcifuge species mean that each has limited flexibility to adapt when displaced from its native soil chemistry. Their restricted distributions therefore represent an outcome of both nutrient availability and the physiological constraints of root system responses.

G. Considering species shared across soil type

The presence of these species in such contrasting soil chemistries indicates that they are not restricted to either calcicole or calcifuge niches. Their persistence can be explained in two ways.

First, some species may exhibit **genetic variation** between populations. In this view, local populations adapt to soil chemistry through allele frequency differences, with variants conferring tolerance to acidic or calcareous conditions favoured at each site. Over time, genetically differentiated ecotypes arise that retain morphological similarity but perform optimally in their respective habitats (Abbott and Robson, 1991).

Alternatively, survival may be underpinned by **phenotypic plasticity**. Here, individuals share similar genetic backgrounds but express different physiological or morphological traits in response to local conditions. For example, variation in root exudates, nutrient uptake strategies, or mycorrhizal associations may be triggered by soil pH, enabling the same genotype to thrive under contrasting chemistries (Schlichting, 1986)

Together, these explanations highlight that species shared across acidic and calcareous soils may persist either through **genetic differentiation among populations** or **plastic responses within individuals**.

Section Three: Birds; species richness across differing habitat types

A. Comparing bird habitats with Sorensen Index

Total species richness for Leighton Moss was 26 species, while 40 species were observed at Castle Head. A comparison of the two sites using the Sorensen Index shows a similarity of 0.42 meaning moderate similarity.

		Castle Head	Leighton Moss	Combined
1	No. Species	26	40	52
2	No. Unique species	12	26	30
3	No. Shared species			14

{Sørensen Index = 2 × shared species / (Castle Head species + Leighton Moss species)
= 2 × 14 / (26 + 40) = 28 / 66 Sørensen Index = 0.42}

Table 8: Calculation of Sorensen index for bird communities observed at Castle Head and Leighton Moss.

The superior species richness at Leighton Moss (RSPB) is likely due to its management to promote bird habitats, including rehabilitated marshes and mudflats that may explain the presence of bitterns and other birds suited wetlands and marshland.

Castle head is however surrounded by farmland and degraded woodlands which would explain lower species richness while allowing for some woodland species to prosper. The overlap points to generalist species such as wrens being suited to a variety of habitats.

B. Comparing bird and plant community similarity across habitat types

The Sørensen similarity index for the grassland plant communities of Scout Scar and Gummer's How was **0.33**, whereas the bird communities of Castle Head and Leighton Moss were more similar, with a value of **0.42**. This contrast highlights important differences in habitat overlap between the two taxa.

The low similarity in plants reflects the strong environmental filtering imposed by soil chemistry. Calcareous limestone at Scout Scar and acidic siltstone at Gummer's How support largely distinct floras, with only a small subset of generalist species shared between them. In contrast, birds are more mobile and less constrained by substrate differences. The moderate similarity between Castle Head and Leighton Moss suggests that while each site supports specialist assemblages shaped by local habitats (woodland versus wetland), there is also a pool of widespread bird species such as wrens and blackbirds capable of exploiting both habitats.

Overall, the comparison illustrates that sessile organisms like plants are tightly bound to edaphic conditions, resulting in sharp community differentiation, while highly mobile taxa such as birds may exhibit broader overlap across contrasting habitats.

Section Four: Analysis of a Limestone Pavement plant community

A. Dispersal vs Woodland Relic hypothesis

The plant communities found in limestone pavement grikes represent a mixture of woodland, grassland and specialist species. These unusual assemblages may reflect persistence of remnant flora, colonisation from surrounding habitats, or both.

Two competing hypotheses attempt to explain the formation of grike flora. The dispersal, or area hypothesis suggests that plant species disperse into grikes from nearby habitats, with larger surface openings capturing more seeds and providing a greater range of microhabitats. In contrast, the remnant, or volume hypothesis proposes that many species are relics of former woodland communities, with grike volume providing buffered microclimatic conditions that enable long-term survival.

To test these hypotheses statistical analysis of data collected at limestone pavement grikes was performed. Spearman's rank correlations were used as species richness is a count variable and may not meet parametric assumptions.

Surface Area and Species Richness Rs = 0.54, $p < 0.001$, n = 60

Volume and Species Richness Rs = 0.42, $p < 0.001$, n = 60

(log-transformed analysis)

The surface area of a grike had a significant positive correlation with species richness (Rs = 0.54, $p < 0.001$, n = 60). A log-transformed regression supported this, showing that surface area explained 29% of variation in richness ($R^2 = 0.29$, slope = 3.17, $p < 0.001$). By contrast, the correlation between grike volume and species richness was also significant but weaker (Rs = 0.42, $p < 0.001$, n = 60), with regression explaining only 15% of variance ($R^2 = 0.15$, slope = 1.60, $p = 0.002$).

Both grike surface area and volume therefore contribute to shaping plant communities, but to different extents. The stronger correlation and higher explanatory power of surface area suggest that colonisation from surrounding habitats is the dominant driver of species richness, supporting the area hypothesis. However, the significant contribution of volume indicates that microclimatic buffering and niche space also play a role, lending partial support to the volume hypothesis alongside the stronger support for the area hypothesis.

R Code used for statistical analysis

```
# Load the grike data
```

```
grike_data <- read_excel("~/Combined field class data 2025 (1).xlsx", sheet = "Limestone Pavement grikes")

grike_data$grike_area_m2 <- grike_data$Grike_width_m * grike_data$Grike_length_m

grike_data$grike_volume_m3 <- grike_data$grike_area_m2 * grike_data$Grike_depth_m

# Spearman's rank correlation for area

cor.test(grike_data$grike_area_m2, grike_data$SpeciesRichness, method = "spearman")

# Linear regression (log-transformed area)

summary(lm(SpeciesRichness ~ log10(grike_area_m2), data = grike_data))

# Spearman's rank correlation for volume

cor.test(grike_data$grike_volume_m3, grike_data$SpeciesRichness, method = "spearman")

# Linear regression (log-transformed volume)

summary(lm(SpeciesRichness ~ log10(grike_volume_m3), data = grike_data))
```

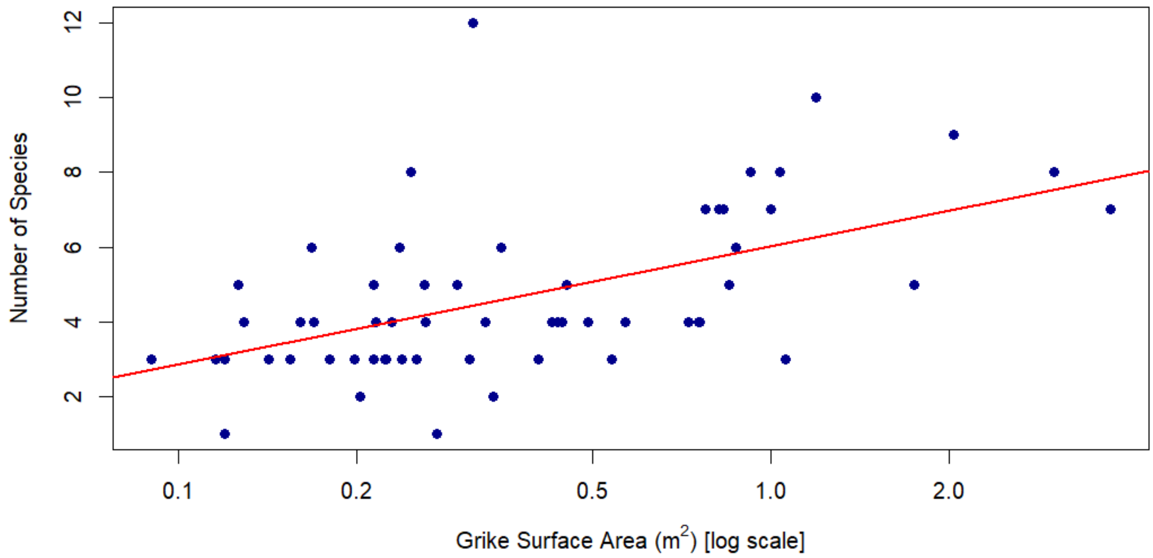

Figure.4: Scatterplot showing the correlation between Grike surface area and number of plant species within, observed with 25x25cm quadrats. A log transformation has been applied on the X-axis to for clarity

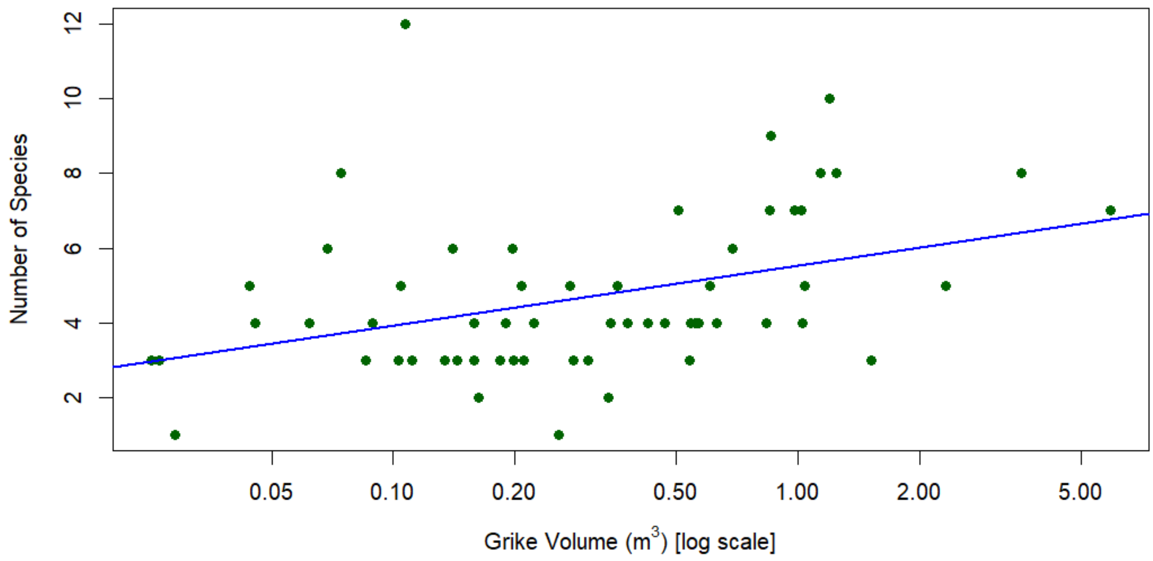

Figure.4: Scatterplot showing the correlation between Grike volume and number of plant species within, observed with 25x25cm quadrats. A log transformation has been applied on the X-axis to for clarity

B. Histograms displaying the rooting depths of select plant species

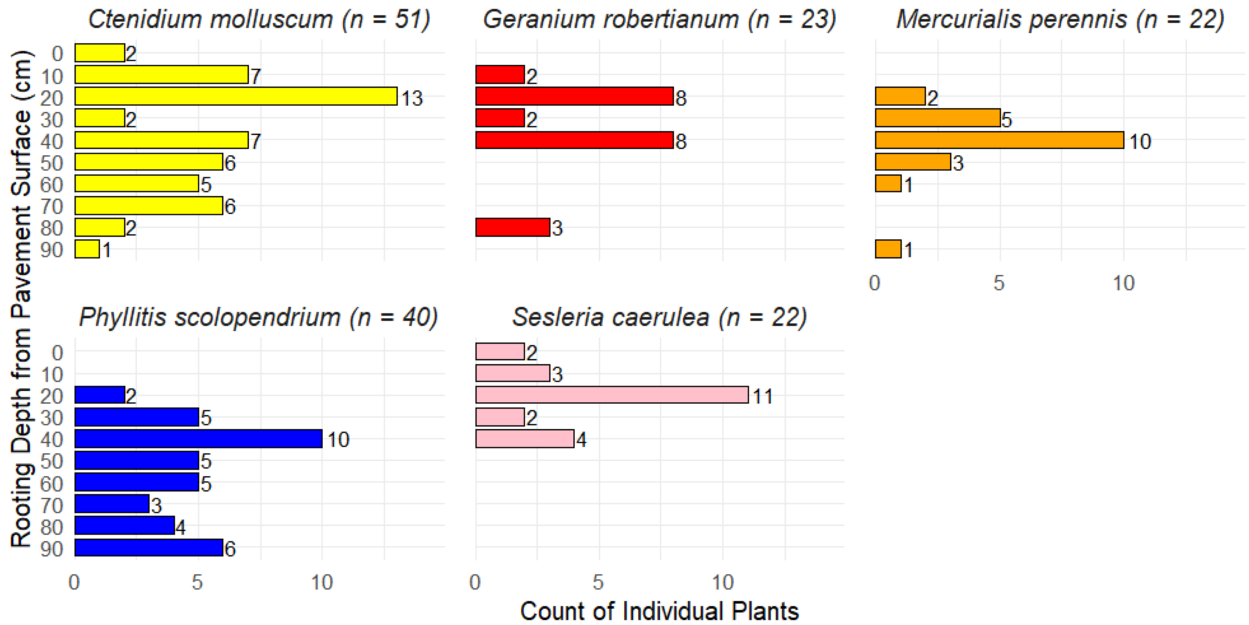

Figure.6: Histograms showing the range of rooting depths for four plant species found on the limestone pavement

C. Analysis of select species rooting depths

Species	**Min Depth (cm)**	**Max Depth (cm)**	**Mean Depth (cm)**
Ctenidium molluscum	1	95	41.4
Geranium robertianum	12	89	38.5

Mercurialis perennis	25	92	43.7
Phyllitis scolopendrium	21	96	59.4
Sesleria caerulea	1	46	25.3

Table 9: Maximum, mean and minimum rooting depths for five selected plant species observed in a limestone pavement habitat.

Phyllitis scolopendrium showed the greatest mean and maximum rooting depths, consistent with its ecology as a shade- and moisture-loving fern adapted to cool, sheltered microhabitats. Mercurialis perennis also rooted relatively deeply, reflecting its typical association with shaded woodland environments where deeper roots aid moisture access and protection from surface disturbance.

Ctenidium molluscum and Geranium robertianum exhibited intermediate rooting depths, indicating ecological flexibility and an ability to occupy a range of positions within grikes. In contrast, Sesleria caerulea was restricted to shallow rooting, reflecting its adaptation to open, sunlit calcareous habitats where it thrives near the pavement surface.

Overall, these results support the interpretation that rooting depth is linked to species' ecological strategies: woodland specialists root more deeply to exploit buffered, moist microclimates, while open-habitat species occupy shallow zones closer to the surface.

D. Most observed species within limestone pavement by preferred habitat type

Preferred Habitat Category	**Main Species (Most observations)**
Woodland	*Mercurialis perennis*
Woodland & other	*Phyllitis scolopendrium*
Walls, Cliffs, Bare Rock	*Ctenidium molluscum*
Grassland	*Sesleria caerulea*
Disturbed Ground	*Senecio jacobaea*

Table 8: Lists species most observed within the limestone pavement using 25x25cm quadrats by their preferred habitat type.

Source: Rodwell (2000), and Averis (2013)

Section Five: Water quality and freshwater invertebrate communities in lentic and lotic waters

A. Freshwater invertebrate communities sampled from lentic (Lake Windemere) and lotic (Cunsey Beck) water.

Lake (Windemere)	*Total*	*% of Sample*	*Stream (Cunsey Beck)*	*Total*	*% of sample*
Gammaridae	122	30.5	Hydracarina	203	36.9
Hydracarina	98	24.5	Gammaridae	202	36.7
Asellidae	44	11	Nemouridae	23	4.2
Hydrobiidae	28	7	Baetidae	21	3.8
Cladocera	24	6	Erpobdellidae	15	2.7

Table 10: Lists the five most numerous Taxa identified from Lake Windemere and Cunsey Beck using three-minute kick sweeps by total individuals and percentage of the total identified individuals.

Taxon Group (Family)	Lake Total	Lake Percent	Stream Total	Stream Percent	BMWP Score
Gammaridae	122	30.5	202	36.7	6
Hydracarina (water mites)	98	24.5	203	36.9	5
Asellidae	44	11	2	0.4	3
Hydrobiidae (Spire Snails)	28	7	0	0	
Cladocera	24	6	0	0	
Baetidae (Swimming MFs)	3	0.8	21	3.8	4
Nemouridae	0	0	23	4.2	7
Ephemeridae (Burrowing MFs)	3	0.8	12	2.2	10
Erpobdellidae	0	0	15	2.7	3
Chironomidae (Non-biting midges)	12	3	0	0	
Caenidae (Square-gilled MFs)	8	2	3	0.5	7
Ceratopogonidae (Biting midges)	6	1.5	5	0.9	
Rhyacophilidae	4	1	6	1.1	7
Ecdyonuridae (Flattened MFs)	1	0.2	9	1.6	10
Corophiidae	4	1	4	0.7	6
Capniidae	0	0	8	1.5	6
Leptoceridae	7	1.8	0	0	
Polycentropidae	3	0.8	4	0.7	7
Gyrinidae (Whirligigs)	1	0.2	6	1.1	5
Phryganeidae	1	0.2	6	1.1	10
Piscicolidae	5	1.2	0	0	
Leuctridae	0	0	5	0.9	10
Leptophlebiidae (Prong-gilled MFs)	4	1	0	0	
Hydrophilidae	3	0.8	1	0.2	5
Dytiscidae (Diving Beetles)	3	0.8	0	0	
Sericostomatidae	3	0.8	0	0	
Simuliidae (Black flies)	3	0.8	0	0	
Glossophonidae	0	0	3	0.5	3
Elminthidae (Riffle Beetles)	2	0.5	0	0	
Lymnaeidae (Pond Snails)	2	0.5	0	0	
Ephemerellidae (Crawling MFs)	0	0	2	0.4	10
Perlidae	0	0	2	0.4	10
Perlodidae	0	0	2	0.4	10
Tipulidae (Crane flies)	0	0	2	0.4	5
Elmidae (Riffle Beetles)	1	0.2	0	0	
Haliplidae	1	0.2	0	0	
Hydroptilidae	1	0.2	0	0	
Odontoceridae	1	0.2	0	0	
Ostracoda	1	0.2	0	0	
Sialidae	1	0.2	0	0	
Hirudidae	0	0	1	0.2	
Hydropsychidae	0	0	1	0.2	5
Philopotamidae	0	0	1	0.2	8
Sub-order Anisoptera (Dragonflies)	0	0	1	0.2	8
Total	400	100	550	100	170

Table 11: Lists Taxa collected at Lake Windemere (Lake) and Cunsey Beck (Stream) using repeated three-minute kick sampling listed by total individuals and as a % of sample. Original BMWP scores listed for Taxa identified in Cunsey Beck

B. Assessment of The BMWP and ASPT Indices

The Biological Monitoring Working Party (BMWP) index is widely used for evaluating freshwater quality by scoring invertebrate families according to their tolerance or sensitivity to pollution. While the initial scoring system was somewhat subjective, it has since been refined and standardised using statically methods informed by large datasets such as the Modified New Walley-Hawkes (MNWH) and Modified Average Score Per Taxon (MASPT) which also accounts for sampling site (Paisley et al., 2013).

A key limitation of the BMWP approach is that scores can artificially increase with greater sampling effort. This drawback is addressed by the Average Score Per Taxon (ASPT) which was introduced alongside the original BMWP system, that accounts for sampling effort by dividing the BMWP score by total Taxa providing a more reliable measure of ecological condition.

The (original) BMWP score for Cunsey Beck (**170**) indicated very high-water quality and when adjusted for total taxa the ASPT score (**6.8**) also indicated very high-water quality.

Cunsey Beck BMWP Score (Table 10) = 170

Number of Taxa = 25

Cunsey Beck ASPT = 6.8

(ASPT = 170/25 = 6.8)

C. Sorensen index for Lake Windemere and Cunsey Beck, and comparison to other Sorensen calculations.

		Lake Windemere	Cunsey Beck	Combined
1	No. Species	31	25	40
2	No. Unique species	17	11	28
3	No. Shared species			14

{Sørensen Index = 2 × shared species / (Lake species + Stream species) = 2 × 14 / (31 + 25) = 28 / 56 Sørensen Index = 0.50}

Table 12: Calculation of Sorensen index for Lake Windemere and Cunsey Beck invertebrate communities.

Sites	Ecological Comparison	Sorensen Index
Lake Windemere vs Cunsey Beck	Freshwater Invertebrates in lentic vs lotic water	0.50
Castle Head vs Leighton Moss	Birds in Woodland vs Marshland	0.42
Castle Hill vs Scout's Scar	Calcareous Plants in Sussex vs Cumbria	0.43
Gummer's How vs Scout's Scar	Plants in acidic vs calcareous soil	0.33

Table 13: Comparison of Sorensen Index calculations across different ecological communities

Cunsey Beck represents a classic lotic (flowing freshwater) ecosystem, characterized by continuous water movement that enhances oxygenation, delivers nutrients, and prevents the accumulation of organic sediments (Malmqvist, 2002). These conditions favour invertebrate taxa adapted to life in flowing water, with many including several identified mayfly taxa displaying streamlined bodies, specialized appendages for clinging to substrates as in Ecdyonuridae, or behaviours such as strong swimming as in Baetidae that help resist the force of the current. Dominant groups in the stream community, also include adaptable taxa such as amphipods (Gammaridae), water mites (Hydracarina) while also including pollution-sensitive insect larvae.

The presence of orders with flattened forms and anchoring mechanisms is typical, and such taxa are often highly susceptible to declines in water quality, explaining their reduced abundance or absence in polluted environments such as Lake Windemere (Harper, 2024).

In contrast, Lake Windermere typifies a lentic (still freshwater) ecosystem, where slower water movement allows sediment and nutrients to accumulate, creating conditions that can range from healthy mesotrophic conditions to more challenging eutrophic conditions. Invertebrate taxa thriving here are generally adapted to lower oxygen and softer substrates, with many exhibiting burrowing habits or filter-feeding strategies to exploit suspended particulates and organic matter. Molluscs and certain crustaceans are numerically dominant, reflecting their tolerance for reduced flow and greater organic enrichment. Some groups found primarily in the lake possess specialized respiratory adaptations or body plans that enable survival in oxygen-poor conditions, such as increased surface area for gas exchange or the ability to utilize atmospheric oxygen. (Harper, 2024)

The Sørensen similarity index for the stream and lake communities is 0.50, indicating moderate overlap. This value is comparable to similarity coefficients observed in bird and calcareous plant communities, suggesting that both physical connectivity and ecological versatility contribute to shared taxa. However, the distinct environmental pressures of lotic and lentic habitats drive clear differences in species composition and relative abundance. The stream supports taxa specialized for high oxygen and mechanical stability, while the lake favors groups adapted to sediment-rich, low-flow conditions.

These patterns reflect the fundamental role of habitat type and environmental adaptation in shaping freshwater invertebrate communities. While hydrological connection and the presence of ecological generalists promote some community overlap, specific adaptations to flow, oxygen availability, and substrate are decisive in determining which groups dominate each site.

Section Five: Terrestrial Insects

A. Analysis of beetle habitat based on size and wing form

Hypotheses concerning average body size

H_0: The average body size of ground beetles does not differ between woodland and grassland.

H_1: The average body size of ground beetles differs between these habitats.

Habitat	**Mean Body Size (mm)**	**SD**	**n**
Woodland	12.9	2.85	160
Grassland	11.3	3.06	156

Table 14: Mean body size (mm), standard deviation (SD), and sample size (n) of ground beetles collected from woodland and grassland habitats at Castle Head

Abundance-weighted mean body size (mm) was calculated for each habitat and habitats were compared using a Welch Two Sample t-test

Welch Two Sample t-test

$t = 4.66$, $df = 311.11$, $p = 4.67 \times 10^{-6}$

95% CI for mean difference: 0.90 to 2.21 mm

There was a statistically significant difference in average beetle size between woodland and grassland habitats ($p < 0.001$). Woodland supported larger-bodied beetle species, consistent with the hypothesis that average body size of ground beetles differs between habitats.

Hypotheses concerning winged, unwinged and dimorphic beetle species habitats

H_0: The frequency distribution of wing forms (winged, unwinged, dimorphic) does not differ between habitats.

H_1: The frequency distribution of wing forms differs between habitats.

Habitat	**Dimorphic**	**Unwinged**	**Winged**
Woodland	3	43	114
Grassland	34	31	91

Table 15: Observed frequency of wing-form of ground beetles collected from woodland and grassland habitats at Castle Head

To test the hypothesis individuals of each wing form per habitat were counted and a Pearson's Chi-squared test of independence was used to compare distributions.

Pearson's Chi-squared test
$X^2 = 30.454$, df = 2, $p = 2.44 \times 10^{-7}$

It was found that wing form distribution is significantly different between habitats ($p < 0.001$). Woodland had more unwinged beetles, while grassland had more dimorphic and winged individuals. This supports the hypothesis that wing forms differ between habitats.

B. Analysis of species richness and expected vs observed habitat of ground beetles

The grassland habitat supported a notably higher ground beetle species richness than woodland, with 16 species recorded compared to only 8 in woodland. This difference is likely driven by the greater structural diversity and microhabitat variety present in grassland, which offers a broader range of resources and niches for beetle species.

Grasslands typically feature open ground, varied vegetation, and fluctuating moisture levels, all of which can support both habitat specialists and generalist species. In contrast, woodland environments often provide fewer microhabitats due to dense canopy cover and thick leaf litter, limiting the diversity of beetles adapted to shaded, stable conditions.

Additional factors such as historical land use, management practices, and levels of disturbance may also contribute to the observed patterns in species richness.

Observed Habitat	**Species**	**Expected Habitat**
woodland	Nebria brevicollis	Woodland/Grassland
woodland	Loricera pilicornis	Woodland/Grassland
woodland	Abax parallelepipedus	Woodland
woodland	Pterostichus madidus	Woodland/Grassland
woodland	Platynus assimilis	Woodland
woodland	Pterostichus nigrita	Woodland/Grassland
woodland	Pterostichus strenuus	Woodland/Grassland
woodland	Bembidion mannerheimii	Grassland
grassland	Abax parallelepipedus	Woodland
grassland	Pterostichus madidus	Woodland/Grassland
grassland	Nebria brevicollis	Woodland/Grassland
grassland	Loricera pilicornis	Woodland/Grassland
grassland	Platynus assimilis	Woodland
grassland	Poecilus versicolor	Grassland
grassland	Clivina fossor	Grassland
grassland	Bembidion lampros	Woodland/Grassland

grassland	Pterostichus nigrita	Woodland/Grassland
grassland	Agonum muelleri	Grassland
grassland	Pterostichus strenuus	Woodland/Grassland
grassland	Bembidion biguttatum	Grassland
grassland	Bembidion guttula	Woodland/Grassland
grassland	Pterostichus anthracinus	Marsh/Fen
grassland	Agonum emarginatum	Marsh/Fen
grassland	Agonum viduum	Marsh/Fen

Table 16: Observed and expected habitat of Carabidae (ground beetle) species collected from Castle Head

Source: Luff (2007)

Habitat	**Generalist Match**	**Exclusive Match**	**Mismatch**
Grassland	7	7	2
Woodland	5	2	1

Table 17: Number of Carabidae (ground beetle) species classified as generalists, habitat matches, or mismatches according to their known habitat preferences, for each surveyed habitat. Generalist species are those typically found in both woodland and grassland; "Match" indicates species found in their preferred habitat, and "Mismatch" indicates species found outside their typical habitat.

Source: from Luff (2007).

Note: Marsh species counted as grassland due to Castle Head's ecology.

Most ground beetle species were found in habitats matching their known preferences from Luff (2007), especially when accounting for generalist species and temporary wet conditions. Species frequently found in marsh or fen habitats were included under grassland due to Castle Head's wet habitat and remaining marshland. Only three species were observed outside their typical habitats, indicating a strong correspondence between literature-based habitat associations and field observations. The identification of generalist species clarifies that many beetles can thrive in multiple habitat types, while mismatches remain rare and likely reflect edge effects or temporary dispersal.

These findings align with Luff's (2007) identified habitat preferences, and provide reporting of generalists and context-sensitive interpretation of marsh/fen specialists in the context of Castle Head.

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